

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 7

In this class we introduce polynomial rings.

1.1. Polynomials. Let R be a ring. A polynomial $f(x)$ in the variable x , and with coefficients in R is a formal sum of the form:

$$(1.1) \quad f(x) = \sum_{n=0}^{\infty} a_n x^n,$$

where $a_n \in R$ and $a_n \neq 0$ for finitely many n . We denote by $R[x]$ the set of all polynomials in the variable x and with coefficients in R . Given two polynomials $f(x), g(x) \in R[x]$:

$$(1.2) \quad f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad g(x) = \sum_{n=0}^{\infty} b_n x^n,$$

we define their sum and product as:

$$(1.3) \quad f(x) + g(x) = \sum_{n=0}^{\infty} (a_n + b_n) x^n,$$

$$(1.4) \quad f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{\substack{k+l=n \\ k,l \geq 0}} a_k b_l \right) x^n.$$

Lemma 1.1. *The set $R[x]$ with binary operations defined in Equations 1.3 and 1.4 is a ring. If R is a ring with unity $1 \neq 0$ then $R[x]$ is a ring with unity, where the unity in $R[x]$ is the constant polynomial 1. If R is a commutative ring then $R[x]$ is a commutative ring.*

Proof. That $(R[x], +)$ is an abelian group is clear since $(R, +)$ is an abelian group (prove this! Note that we are just doing addition in R separately for each degree). It remains to prove that multiplication in $R[x]$ is associative and that multiplication in $R[x]$ satisfies the distributive law. Associativity is proved in the book, so we will

prove that the distributive law holds. Let $f(x), g(x), h(x) \in R[x]$ be polynomials with coefficients:

$$(1.5) \quad f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad g(x) = \sum_{n=0}^{\infty} b_n x^n, \quad h(x) = \sum_{n=0}^{\infty} c_n x^n.$$

Using Equations 1.3 and 1.4 we calculate:

$$\begin{aligned} f(x)(g(x) + h(x)) &= \left(\sum_{n=0}^{\infty} a_n x^n \right) \cdot \left(\sum_{n=0}^{\infty} (b_n + c_n) x^n \right) = \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} a_k (b_l + c_l) \right) x^n = \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} (a_k b_l + a_k c_l) \right) x^n = \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} a_k b_l + \sum_{k+l=n} a_k c_l \right) x^n = \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} a_k b_l \right) x^n + \sum_{n=0}^{\infty} \left(\sum_{k+l=n} a_k c_l \right) x^n = \\ &= f(x)g(x) + f(x)h(x) \end{aligned}$$

which proves that the distributive law holds.

Assume now that the ring R has a unity $1 \in R$. Let $\delta_{0k} = 1$ if $k = 0$ and $\delta_{0k} = 0$ if $k \neq 0$. We can write the constant polynomial 1 in $R[x]$ as:

$$(1.6) \quad \sum_{n=0}^{\infty} \delta_{0n} x^n$$

and if we multiply this polynomial with any other polynomial then:

$$\begin{aligned} \left(\sum_{n=0}^{\infty} \delta_{0n} x^n \right) \cdot \left(\sum_{n=0}^{\infty} a_n x^n \right) &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \delta_{0k} a_l \right) x^n = \\ &= \sum_{n=0}^{\infty} \left(1 \cdot a_n + \sum_{\substack{k+l=n \\ k \geq 1}} \delta_{0k} a_l \right) x^n = \\ &= \sum_{n=0}^{\infty} a_n x^n \end{aligned}$$

so the constant polynomial 1 is indeed a unity in $R[x]$. We leave the proof of commutativity as an exercise. ■

Remark 1. Let δ_{mn} be defined as $\delta_{mn} = 1$ when $m = n$ and $\delta_{mn} = 0$ when $m \neq n$. If R is a ring with unity then we can define polynomials:

$$(1.7) \quad x^m = \sum_{n=0}^{\infty} \delta_{mn} x^n.$$

We leave it as an exercise to check that with this description of x^m one obtains $x^n x^m = x^{n+m}$. Combining this with the distributive law justifies doing 'normal' manipulations with polynomials whenever we want to multiply them.

Definition 1.1. If R is a ring, then $R[x]$ is the polynomial ring with unknown variable x and with coefficients in R .

Remark 2. We can also define polynomials in two variables by taking the polynomial ring of a polynomial ring. Indeed if R is a ring then $R[x]$ is also a ring, and we can define the polynomial ring in two variables as $R[x, y] = (R[x])[y]$.

Theorem 1.1. Let F and E be fields such that F is a subfield of E , $F \leq E$. If $\alpha \in E$ then the map:

$$(1.8) \quad \phi_\alpha : F[x] \rightarrow E, \quad \phi_\alpha(a_n x^n + \dots + a_0) = a_n \alpha^n + \dots + a_0,$$

is a homomorphism of rings that satisfies $\phi_\alpha(x) = \alpha$, $\phi_\alpha(a) = a$ for $a \in F$.

Remark 3. The map ϕ_α is the evaluation homomorphism at $\alpha \in E$.

Proof. We begin by showing that ϕ_α is a homomorphism. First, note that:

$$\begin{aligned} \phi_\alpha \left(\sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} b_n x^n \right) &= \phi_\alpha \left(\sum_{n=0}^{\infty} (a_n + b_n) x^n \right) = \\ &= \sum_{n=0}^{\infty} (a_n + b_n) \alpha^n = \\ &= \sum_{n=0}^{\infty} (a_n \alpha^n + b_n \alpha^n) = \\ &= \phi_\alpha \left(\sum_{n=0}^{\infty} a_n x^n \right) + \phi_\alpha \left(\sum_{n=0}^{\infty} b_n x^n \right) \end{aligned}$$

so ϕ_α respect addition. To see that ϕ_α respects multiplication we calculate similarly:

$$\begin{aligned}
 \phi_\alpha \left(\left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{n=0}^{\infty} b_n x^n \right) \right) &= \phi_\alpha \left(\sum_{n=0}^{\infty} \left(\sum_{k+l=n} a_k b_l \right) x^n \right) = \\
 &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} a_k b_l \right) \alpha^n = \\
 &= \sum_{n=0}^{\infty} \sum_{k+l=n} (a_k \alpha^k) (b_l \alpha^l) = \\
 &= \left(\sum_{n=0}^{\infty} a_n \alpha^n \right) \left(\sum_{n=0}^{\infty} b_n \alpha^n \right) = \\
 &= \phi_\alpha \left(\sum_{n=0}^{\infty} a_n x^n \right) \phi_\alpha \left(\sum_{n=0}^{\infty} b_n x^n \right)
 \end{aligned}$$

which shows that ϕ_α respects multiplication. We leave the last two claims as an exercise. ■

Definition 1.2. Let $f(x) \in F[x]$ be a polynomial with coefficients in F . If $F \leq E$ and $\alpha \in E$ then we say that $f(x)$ has a zero at α if $\phi_\alpha(f(x)) = 0$.

Email address: sandfeldt@uchicago.edu